

Research Report: Healthcare Data Insights and Predictive Modeling

1. Introduction

Healthcare data science supports evidence-based decision-making by analyzing patient records and identifying patterns associated with outcomes. This project applies an end-to-end analytics and machine-learning workflow to predict early hospital readmission among diabetic patients.

2. Dataset and Prediction Task

The selected dataset is the Diabetes 130-US Hospitals for Years 1999-2008 dataset. It contains 101,766 hospitalized patient records and 47 features in the UCI version. The prediction task is binary classification: whether a patient is readmitted within 30 days of discharge. The UCI repository identifies classification as an associated task and notes that the dataset contains missing values. The dataset covers clinical care at 130 US hospitals and integrated delivery networks from 1999–2008.

3. Exploratory Data Analysis

EDA examines the dataset structure, missing values, target distribution, demographics and relationships between important features and early readmission. Visualizations include target-distribution plots and feature-versus-target comparisons.

4. Data Cleaning and Preparation

The source dataset uses '?' to represent missing values. These are converted to missing values and handled through preprocessing. Encounter and patient identifiers are excluded from modeling. Numerical variables are median-imputed and standardized; categorical variables are mode-imputed and one-hot encoded.

5. Handling Class Imbalance

Early readmission is a minority-class outcome. SMOTE is used inside the training pipeline so that synthetic oversampling occurs only after the training split, reducing the risk of test-set leakage.

6. Classification Models

Two core models from the internship brief are implemented: Logistic Regression and Random Forest. Logistic Regression provides an interpretable baseline, while Random Forest can capture nonlinear relationships.

7. Evaluation

The models are evaluated using accuracy, precision, recall, F1-score, ROC-AUC and confusion matrices. In healthcare prediction, minority-class recall and F1-score are especially important because missing a high-risk patient can be more consequential than a false positive.

8. Responsible Use and Limitations

The dataset is historical and may not represent current clinical practice. Demographic information can be sensitive. Predictions should be treated as analytical decision support rather than medical diagnosis. External validation and clinical oversight would be necessary for real-world deployment.

9. Expected Contribution

The project demonstrates data cleaning, EDA, preprocessing, imbalance handling and supervised classification in a healthcare context. The final notebook will provide reproducible metrics and visualizations for comparing the two models.